

NSPL 0.5 μ m Analog CMOS 2P3M 5V General Purpose Input/Output

Version 1.0

National Semiconductor Public Laboratory (NSPL)

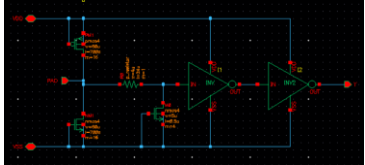
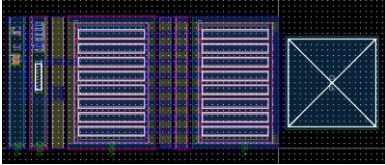
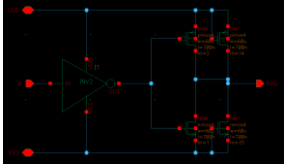
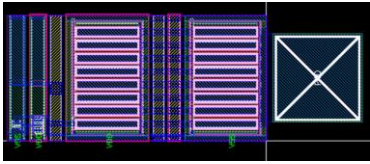
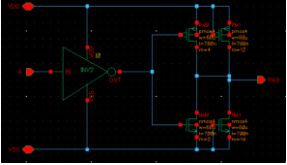
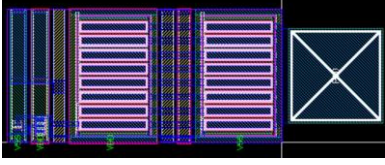
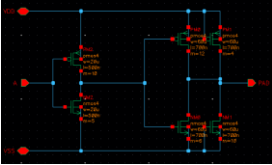
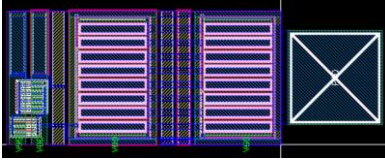
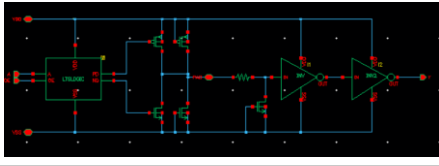
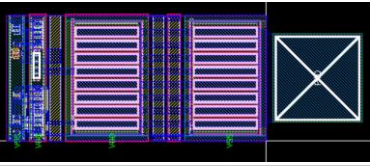
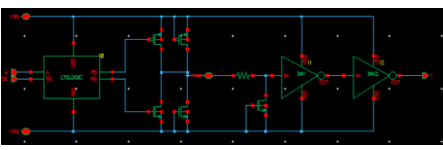
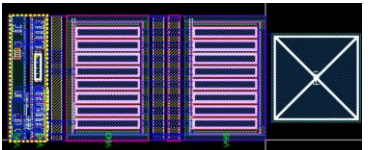
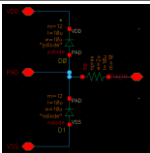
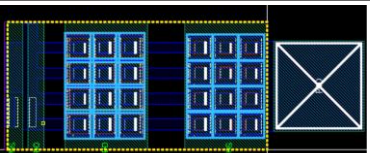
- Electronics and Telecommunications Research Institute (ETRI)
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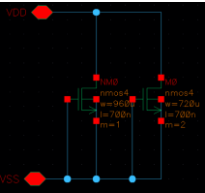
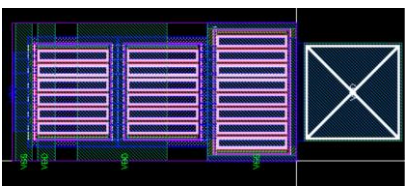
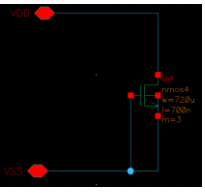
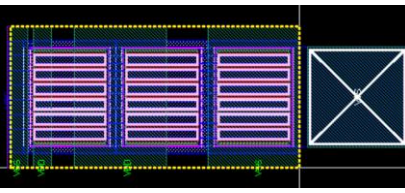
서울대학교
SEOUL NATIONAL UNIVERSITY



❖ 5V GPIO Cell List(1/2)

Cell List	Purpose	Schematic	Layout
PIC	Digital Input Cell		
POB4	Output Cell with 4 mA Driving Ability		
POB8	Output Cell with 8 mA Driving Ability		
POB24	Output Cell with 24 mA Driving Ability		
PBCT4	Bi-directional Cell with 4 mA Driving Ability, CMOS Input		
PBCT8	Bi-directional Cell with 8 mA Driving Ability, CMOS Input		
PANA	Analog Cell (N/P Diode Size : $30\text{ }\mu\text{m} \times 30\text{ }\mu\text{m} \times 4$)		

❖ 5V GPIO Cell List(2/2)

Cell List	Purpose	Schematic	Layout
PVDD	Power Clamp Cell		
PVSS	Ground Clamp Cell		
PCORNER	Corner Cell	-	
IOFILLER50	Filler Cell	-	
IOFILLER10	Filler Cell	-	
IOFILLER1	Filler Cell	-	

❖ Remarks

1. 모든 IO Cell은 목적에 맞게 사용되어야 함 (목적과 다른 IO 사용시 칩 오동작 또는 Fail 발생 가능)
2. 사용하는 Cell은 구동을 위해 PVDD와 PVSS와 전기적 연결 필요(IO Cell간 MET3로 직접 배선, 또는 Filler 사용)
3. 신호 포트는 MET2를 이용하여 내부 회로와 배선 연결
4. PANA Cell은 저항이 포함된 Port와(PADR) 저항이 없는 Port가(PAD)있음
→ 저항이 포함되지 않는 Port를 사용할 경우 사전 협의 혹은 설계자가 내부 회로 ESD Fail에 대해 고려
5. POB24 Cell은 Drivability 가 큰 Cell로 꼭 필요할 경우에 한해 Max. Frequency & PAD Loading조건 등을 Simulation으로 확인 후 사용 요망
6. 측정 평가 시, PANA를 제외한 입력 PAD 및 미사용 IO PAD는 Floating 상태로 두지 말것 (신호 인가 없을 시 큰 누설 전류 발생 가능) → High 또는 Low 전위 인가 필요 (단, 출력 전용 PAD는 High/Low를 강제로 인가하지 않도록 주의)
7. 내부회로가 Metal1 또는 Metal2 까지만 공정이 진행되더라도, 모든 PAD에는 (MET1, Via1), MET2, Via2, MET3, PAD Layer 반드시 필요